UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

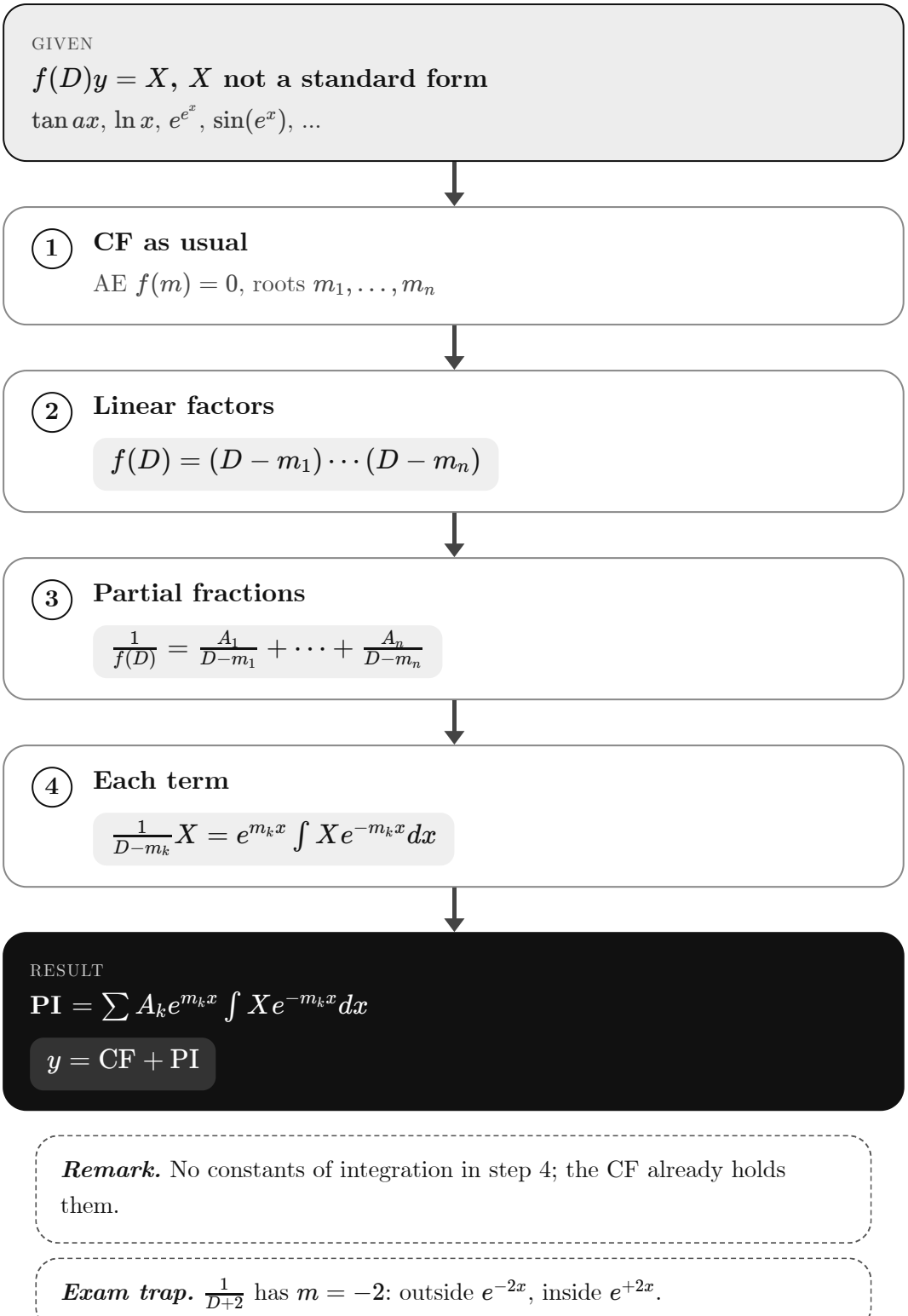
Class 009 · Particular integral when X is any other form · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S009·5	Unit 2 deck 009 (“Particular integral when X is any other form”, SESSION 7), slide 5 (rendered .pptx)
T009 00:03:13	Unit 2 AV-summary transcript “Particular Integral when X is any other form” (15.2 min), segment start
T009b 00:01:10	Unit 2 AV-summary transcript “Particular integral when X is any other form_Continued” (10.2 min), segment start
CW U2 P28	Unit 2 class-work problem 28 (pointer only; solved in the class 011 VoP pack)
ESA Dec-23 3c	Past paper: UE22MA141A Dec 2023, question 3.c

Contents

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2. Figure F1.2 · Partial fractions or one factor at a time?
3. Figure F2.1 · Slides 009·5–7: $(D^2+a^2)y=\tan ax$
4. Figure F3.1 · ESA Dec-23 3c: $y''+3y'+2y=e^{e^x}$
5. Figure F3.2 · Slides 009·14–17: $(D+2)(D+3)y=e^{2x}\sin(e^x)$
6. Which PI rule? (the four standard forms and “any other form”)
7. $\tan$, $\sec$, $\csc$ right-hand sides: operator method or variation of parameters
8. Every “any other form” problem in this pack

1 The general method: linear factors, partial fractions and $\frac{1}{D-m}X$

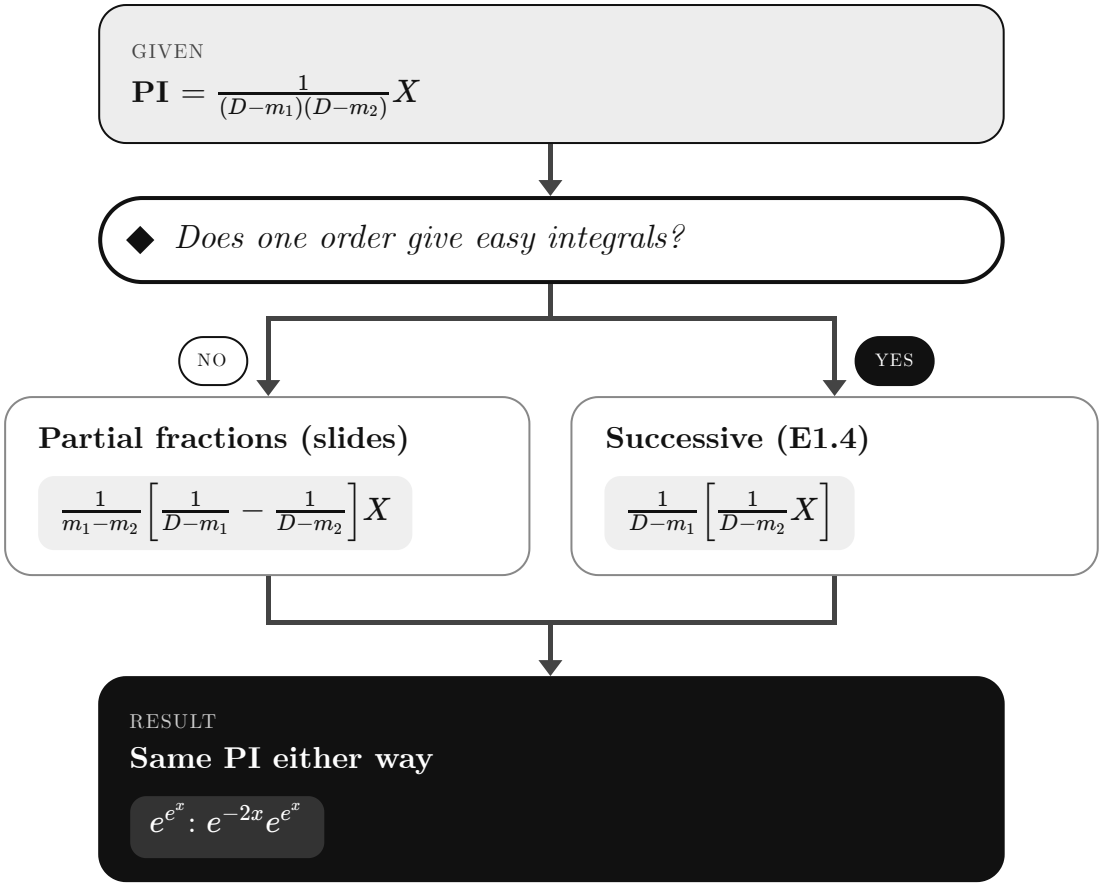


Remark. No constants of integration in step 4; the CF already holds them.

Exam trap. $\frac{1}{D+2}$ has $m = -2$: outside e^{-2x}, inside e^{+2x}.

Figure F1.1. PI when X is any other form (slides 009·3–4) ESA ★★

S009·3 · S009·4 · T009 00:00:58 · T009 00:01:32 · T009 00:02:11



Remark. The slides always use partial fractions; the successive route is a check or a shortcut. For a repeated factor $(D-m)^2$ use the successive route.

Figure F1.2. Partial fractions or one factor at a time? ESA ★★

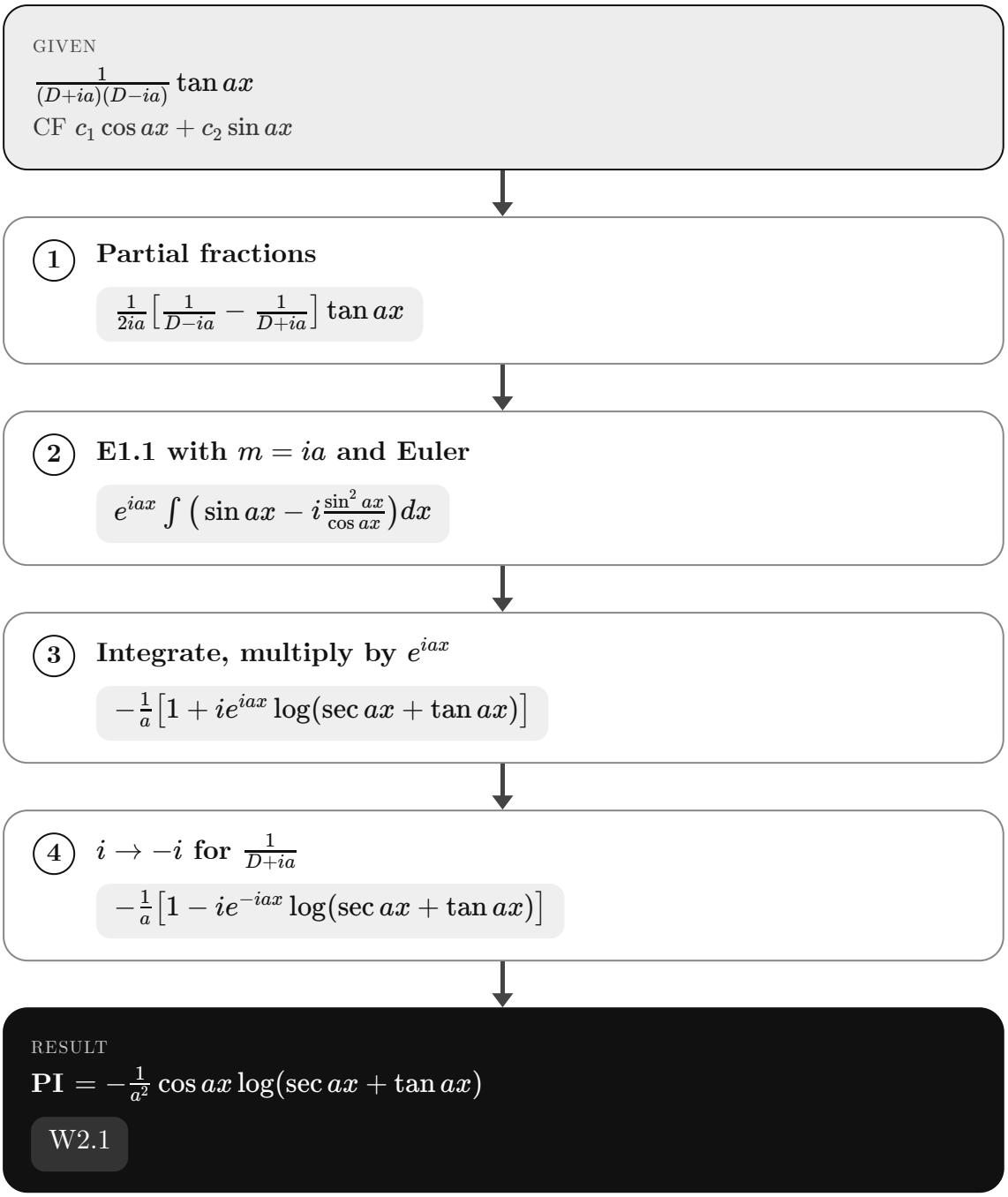
ADDED FOR CLARITY S009·4 · S009·10 · ESA Dec-23 3c

Which PI rule? (the four standard forms and "any other form")

X	Rule	Class pack
e^{ax}	$D \to a$	003
$\sin(ax+b)$, $\cos(ax+b)$	$D^2 \to -a^2$	004
x^m	series in D up to D^m	005
$e^{ax}V$ **(and** xV**)**	$e^{ax}\frac{1}{f(D+a)}V$; $\left[\frac{x}{f(D)} - \frac{f'(D)}{(f(D))^2}\right]V$	006
anything else	linear factors, partial fractions, $\frac{1}{D-m}X = e^{mx}\int Xe^{-mx}dx$	009 (this pack)

T009 00:00:27 · T009b 00:00:32 · S009·3

2 Complex linear factors: $(D^2 + a^2)y = \tan ax$



Remark. The constants $-\frac{1}{a}$ cancel and $e^{iax} + e^{-iax} = 2\cos ax$ makes the result real.

Figure F2.1. Slides 009·5–7: $(D^2 + a^2)y = \tan ax$ ESA ★★

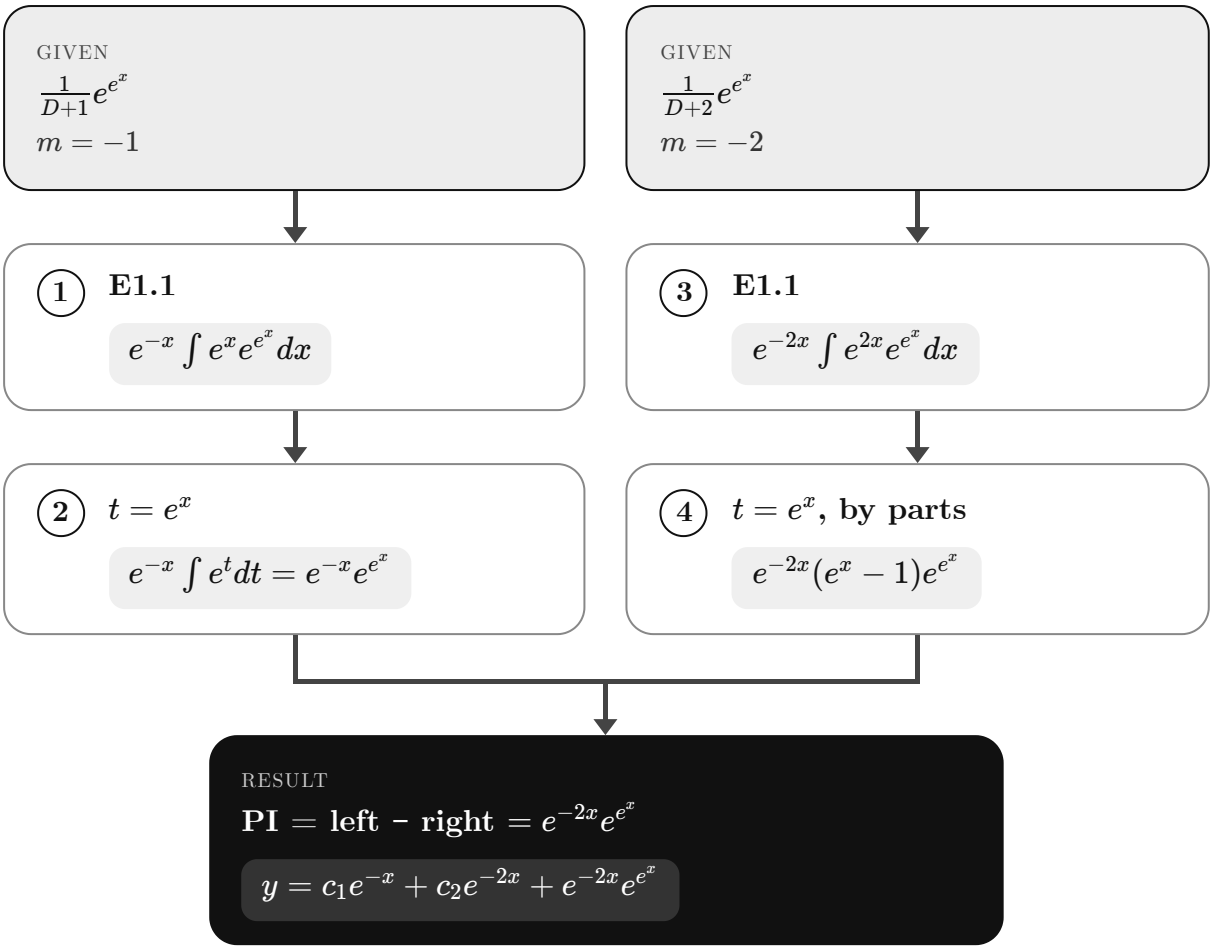
S009·5 · S009·6 · S009·7 · T009 00:04:12 · T009 00:11:09

tan, sec, csc right-hand sides: operator method or variation of parameters

	Operator method (this pack)	Variation of parameters (class 011)
Idea	$\frac{1}{D-m}X = e^{mx}\int Xe^{-mx}dx$, complex $m = \pm ia$	$y_p = -y_1\int\frac{y_2X}{W}dx + y_2\int\frac{y_1X}{W}dx$
Arithmetic	complex numbers, Euler's formula	real; needs the Wronskian W
Same answer?	yes: $\frac{\tan ax}{D^2+a^2} = -\frac{1}{a^2}\cos ax\log(\sec ax + \tan ax)$	yes (up to CF terms)
Asked as	slides 009·5–7; ESA Dec-23 3c (e^{e^x})	CW P23 ($\sec ax$), P26 ($\csc x$), P28 ($\tan 2x$); ESA VoP questions
When to use	question says "solve" and X is no standard form	question says "by the method of variation of parameters"

ADDED FOR CLARITY S009·5 · S009·7 · CW U2 P23 · CW U2 P26 · CW U2 P28 · ESA Jan-24 s1 3a

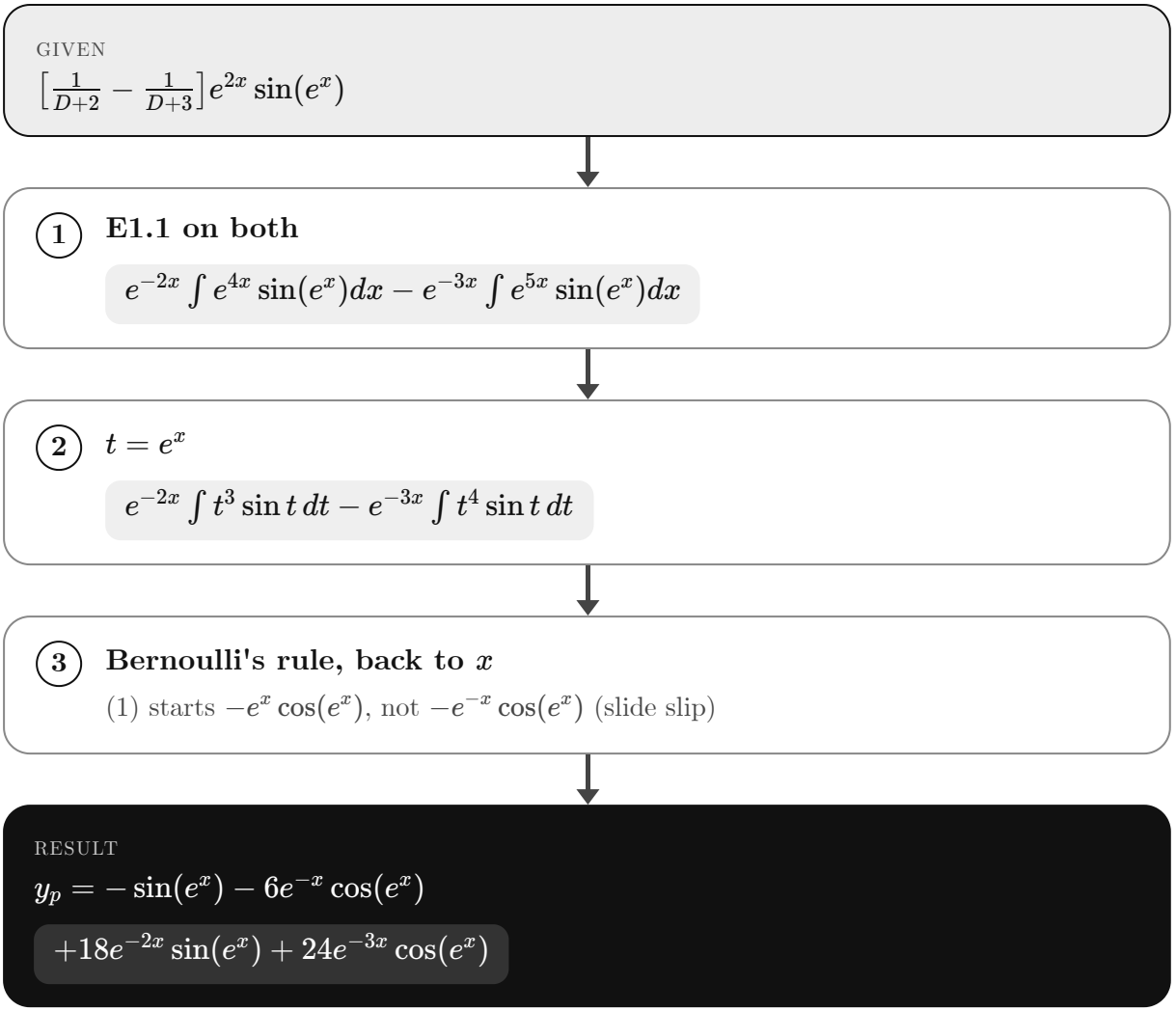
3 Right-hand sides in e^x: the substitution $t = e^x$



Remark. CF from $m = -1, -2$. Worked in full as W3.1 (7 marks).

Figure F3.1. ESA Dec-23 3c: $y'' + 3y' + 2y = e^{e^x}$ ESA ★★

ESA Dec-23 3c · S009·10 · S009·11



Exam trap. Slide 009·16's line (1) has $-e^{-x}\cos(e^x)$; with that slip the e^x terms would not cancel.

Figure F3.2. Slides 009·14–17: $(D+2)(D+3)y = e^{2x}\sin(e^x)$ ESA ★★

S009·14 · S009·15 · S009·16 · S009·17

Every "any other form" problem in this pack

Source (worked)	Equation	Integrals after $t = e^x$ / by parts	PI
S009·8–9 (W1.1)	$D^2y = \ln x$	$\iint \ln x$	$\frac{x^2 \ln x}{2} - \frac{3x^2}{4}$
S009·5–7 (W2.1)	$(D^2 + a^2)y = \tan ax$	$\int \sin ax$, $\int \sec ax$	$-\frac{1}{a^2}\cos ax \log(\sec ax + \tan ax)$
S009·10–11, ESA (W3.1)	$(D^2 + 3D + 2)y = e^{e^x}$	$\int e^t$, $\int te^t$	$e^{-2x}e^{e^x}$
S009·12–13 (W3.2)	$(D + 1)(D + 2)y = \sin(e^x)$	$\int \sin t$, $\int t \sin t$	$-e^{-2x}\sin(e^x)$
S009·14–17 (W3.3)	$(D^2 + 5D + 6)y = e^{2x}\sin(e^x)$	$\int t^3 \sin t$, $\int t^4 \sin t$	$-\sin(e^x) - 6e^{-x}\cos(e^x) + \dots$

ADDED FOR CLARITY S009·5 · S009·8 · S009·10 · S009·12 · S009·14 · ESA Dec-23 3c